

Chapter 11

Advanced Exercises

The beginner's program described in chapter 10 should be sufficient for daily spine health. Several situations, however, may call for further training:

- Certain occupational tasks with specific demands that require unique preparation and training
- Athletic endeavors that demand higher challenges of low back training (although this is achieved with much higher risk of tissue damage from overload)
- Some patients who, as they progress with the isometric stabilization exercises described in chapter 10, want to continue increasing the challenge

Athletic performance, or simply increasing ability, may be enhanced with specific exercises. However, performance can also be ruined by exercises that cause imbalances between performance variables, and injury. Joints need more resilience when called upon to bear more load. Speed must be developed along with an enhanced ability to steer the forces through the body linkage. But what are the variables that increase leg speed or the speed of a tennis racket? There are many, including enhanced core stiffness to increasing distal limb speed. Mobility at the distal joints needs tuning. Creating strength in pulses enhances limb speed, which implies that training the rate of muscle force development and relaxation is important (McGill et al., 2010). Studies linking specific fitness variables with performance can be surprising, such as those we have conducted with occupational groups (McGill, Frost, and Crosby, 2013; McGill et al., 2013) and sporting groups (e.g., McGill, Andersen, and Horne, 2012). Choosing the best exercise tool to achieve each performance while considering the person's current level of competency, injury history, and body type is not a trivial task; it requires a great degree of knowledge and skill.

We have quantified the spine loading and athleticism developed during strongman tasks (McGill, McDermott, and Fenwick, 2008), kettlebell swinging (McGill and Marshall, 2012), hula hooping (McGill and Andersen,

2015), powerlifting (Cholewicki, McGill, and Norman, 1991), and mixed martial arts (McGill et al., 2010), to name just a few. All of these approaches have a place but the best program for an individual is focused. Our case studies of the great athletes revealed that the best performers were rarely those who tested as the strongest. The best performers, rather, had superior technique. They listened to their bodies and preserved their joints. Many used quite novel techniques to enhance the full spectrum of motor skill and strength and speed variables in planned programs. These programs cleverly scheduled rest periods alongside enhanced restoration techniques. The successful athletes worked on the aspects that would create the largest enhancement in performance.

Finally, we are not born equal. Many studies have documented the genetically determined differences in response to training. There are both responders and nonresponders. A technique that enhances the squat performance in one person may have no effect on another. The second person using that technique has wasted effort that could have been directed in a more productive way. This is not to discourage program design for enhancing performance, but rather to illustrate that working with an athlete with back pain to become a champion requires every bit of knowledge I have. With each of my athlete consulting cases, I proceed realizing that I will need to follow principles, but the best approach is not always obvious.

This chapter briefly introduces a few examples of how to increase low back challenges safely, addresses specific worker and athletic concerns regarding advanced training, and provides some ideas about the next steps to take in the study of low back rehabilitation and training. The goal of this book is to help clinicians and people with back disorders reduce pain and enhance function. However, advanced training is a complex topic that cannot be adequately addressed in a single chapter. See *Ultimate Back Fitness and Performance* (McGill, 2014) for a detailed treatment of advanced exercise for the back.

Chapter 10 covers exercises that accomplish the goals of the first three stages of progressing exercise. This chapter introduces a few notions about stages 4 and 5. Following is a recap of the stages as introduced in chapter 8:

- Stage 1: Establish quality movement engrams (motion patterns, motor patterns, and corrective exercise).
 - Identify perturbed patterns and develop appropriate corrective

- exercise.
- Address basic movement patterns through to complex-activity patterns.
- Address basic balance challenges through to complex and specific balance challenges.
- Stage 2: Build whole-body and joint stability (focus on spine stability here).
 - Build stability while sparing the joints.
 - Ensure sufficient stability commensurate with the demands of the task.
 - Apply these patterns to the performance of daily activities.
- Stage 3: Increase endurance.
 - Address basic endurance training to ensure the capacity needed for stabilization.
 - Address activity-specific endurance (duration, intensity).
 - Increase the ability to repeat pristine movement patterns without fatigue-induced compromise.
 - Build the base for eventual performance training (only in those with this goal).
- Stage 4: Build strength.
 - Spare the joints while maximizing the neuromuscular compartment challenge.
 - Progress to skill movements.
- Stage 5: Develop speed, power, and agility.
 - Develop ultimate performance based on the foundation laid in stages 1 through 4.
 - Focus on optimizing elastic energy storage and recovery.
 - Employ the techniques of superstiffness. Stiffen the core and unleash the athleticism of the hips, shoulders, and distal joints.

Safely Increasing Challenges

To optimize patient safety, you must be aware of issues associated with training on labile surfaces and with machines, and how to progress the big three exercises described in chapter 10 (curl-up, side bridge, and bird dog) to their highest level. Some guidance is provided here.

Labile Surfaces and Resistance Training Machines

Not all clinicians are aware of the factors they should take into account when deciding whether to prescribe exercise using labile surfaces or machines. Following are guidelines about both of these.

- Training with labile surfaces. Challenges to the spine during daily activity include maintaining stability during static, steady-state postures, unexpected loading events, and planned dynamic or ballistic movement. This has motivated some clinicians to recommend exercising on labile surfaces such as gym balls and suspension straps. Certainly, these labile surfaces challenge the motor system to meet the dynamic tasks of daily living or specific athletic activities and can be very helpful for advanced training. But might this type of training be of concern for some patients? Our recent quantification of elevated spine loads and muscle coactivation when performing a curl-up on labile surfaces (Vera-Garcia, Grenier, and McGill, 2000) suggests that the rehabilitation program should begin on stable surfaces. In this case, we assessed the simple curl-up for the effect of a labile surface on muscle activation patterns (see [figure 11.1](#), a-d). Simply moving from a stable surface to a labile surface caused much more cocontraction, which in many cases virtually doubled the spine load (see [figure 11.2](#)). The practice of placing patients on labile surfaces early in the rehabilitative program can delay improvement by causing exacerbating spine loads. We therefore suggest beginning exercises on a stable surface and establishing a positive slope to improvement. Introduce labile surfaces judiciously only once the patient has achieved spine stability and sufficiently restored load-bearing capacity, and can tolerate additional compression. This principle

can be extended to sitting. Sitting on a gym ball greatly elevates spine load through increased muscle coactivation. For this reason, those without back issues should avoid prolonged sitting on gym balls, and patients should use them only once they have achieved spine stability and increased load-bearing capacity.

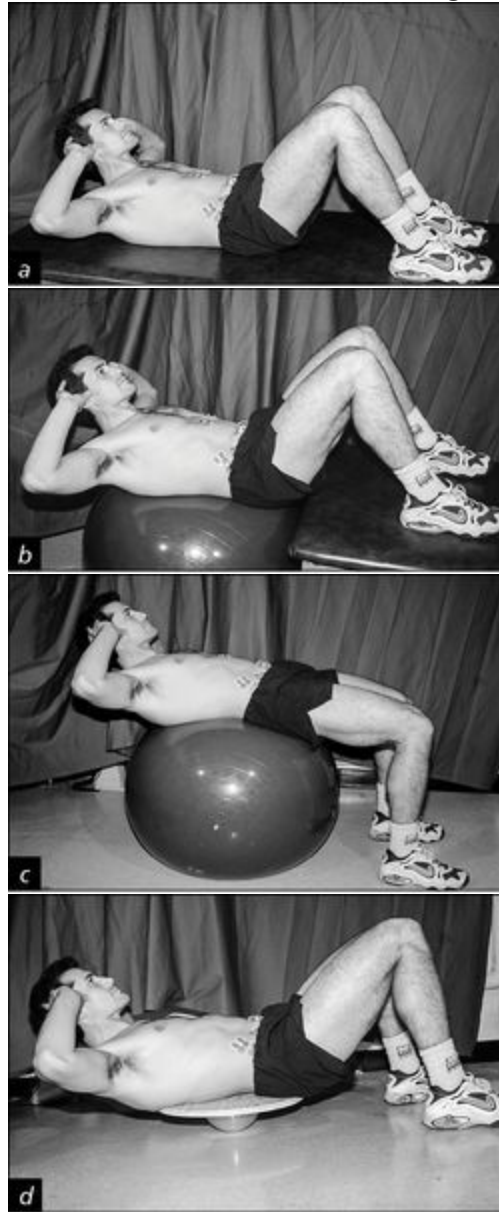
There is a time and a place for labile surfaces. Pushing and pulling exercises need to be incorporated into any advanced back conditioning program. Comparing pushes and pulls on stable surfaces and with suspension straps documents the additional muscle challenges and corresponding spine and joint loads (McGill et al., in press a and b). More coordination is needed to control the direction of force application with labile surfaces, which is an essential skill. The cost is that labile surfaces limit the brute strength developed because the weakest component defines the balance of the other joint axes (which may also be an asset). Joint loading is also higher as a result of the extra cocontraction of many muscles needed for control.

Training with machines and equipment. Generally, the goal of establishing stabilizing motor patterns requires the person to support body weight and coordinate the stabilization of all joints involved in the task. In other words, it is a whole-body, even whole-person, endeavor. Many training machines, on the other hand, are made to isolate a specific joint. Because neither workers nor athletes perform their tasks in this artificially stabilized manner, these types of motor patterns may not be transferable and, worse, may cause inappropriate motor and motion patterns. The only instances in which machines that isolate joints can be helpful is when an injury to a specific body part requires its protection during rehabilitative training. Some incorrectly believe that isolating a joint somehow reduces the loading and thus reduces the risk. Consider, for example, the typical extensor benches or Roman chair, which generally impose twice the muscle loading than results from performing the bird dog, because both the right and left sides of the lumbar and the thoracic extensors are activated (see [figure 11.3](#)). Patients who must use machines should consider certain back-sparing techniques. Ensuring that all movement occurs about the hip joint, rather than the spine, would be an example of such a technique.

For the purposes of this discussion, cables to weight stacks are not considered machines because they do not isolate joint motion and add

resistance to whole-body motion patterns. One exercise that enhances the ability to stabilize the spine is the cable pull-down. This exercise, when the handlebar is brought down to the chest (rather than the back, which is the traditional technique), challenges the latissimus dorsi and other stabilizers.

Figure 11.1 The simple curl-up was assessed for the effects of a labile surface on muscle activation patterns. The percentage of maximal voluntary contraction (%MVC) in the rectus abdominis and external and internal obliques caused by simply moving from (a) a stable surface to (b-d) varying types of labile surfaces is shown in figure 11.2.



Photos from Stuart McGill

Figure 11.2 The %MVC caused by each of the postures shown in figure 11.1. A curl-up with the body over a ball and the feet on the floor (figure 11.1c) virtually doubles the abdominal muscle activation of a curl-up on a stable surface (figure 11.1a) and, correspondingly, the spine load. Note that the %MVC required of the three muscles studied is also much higher in curl-ups with the body over a ball and the feet on a bench (figure 11.1b) and with the body on a wobble board (figure 11.1d) rather than on a stable surface.

Clearly, a gym ball can be wonderful for advanced training but is contraindicated for many patients.

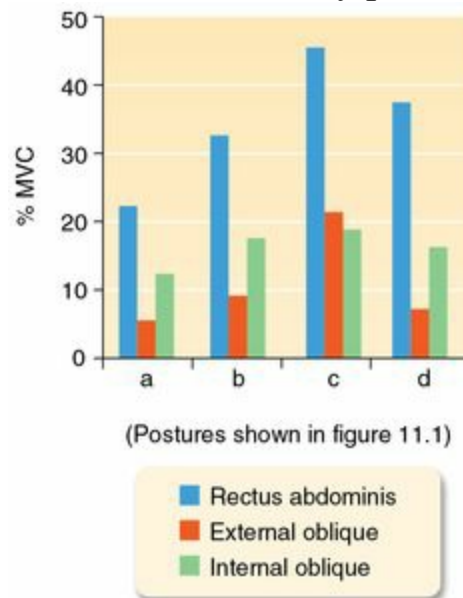


Figure 11.3 Typical back extensor devices impose generally twice the muscle loading of the bird dog because both the right and left sides of the lumbar and the thoracic extensors are activated. These may be necessary for some types of performance training, but they are ill-advised for most patients because of the spine compression and motion. Enhancing strength for people without pain would be wiser if they eliminated the spine movement when under load, and rotated about the hips.



Photos from Stuart McGill

Safely Progressing Back Exercises

The curl-ups, side bridges, and bird dogs pictured in chapter 10 can be made more challenging and yet still incorporate the safety features noted. After explaining these advanced forms of the big three exercises, I present an additional advanced back exercise.

Advanced Back Exercises

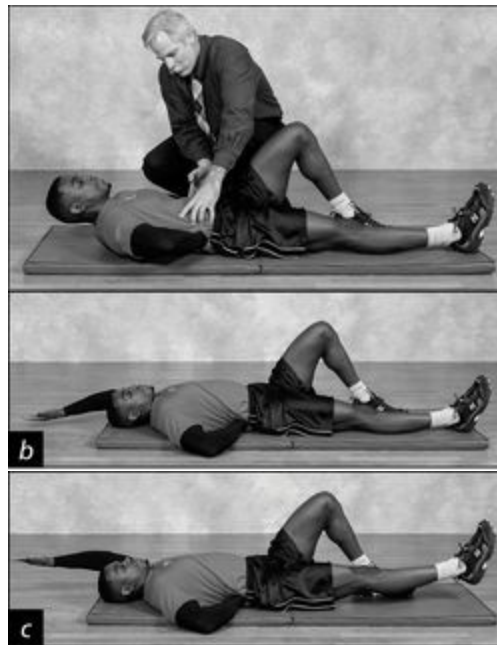
Once the patient has mastered the advanced curl-up, side bridge, and bird dog described in chapter 10, instruction to try the following may be appropriate.

Curl-Up, Highest Level

1. The patient braces the abdomen (if necessary, review the instructions).
2. The patient curls up against the brace, but not any higher than in other

forms of the exercise.

3. In the up position and while maintaining bracing, the patient takes a few deep breaths (a). This level of curl-up would be challenging for the toughest NFL linebacker!
4. The abdominal wall is cued for bracing and stiffening. Plyometrics may then be added in the dead bug position (b, c) with short-range but rapid hip and shoulder flexion. All motion must occur in the hip and shoulder joints, not in the braced spine.



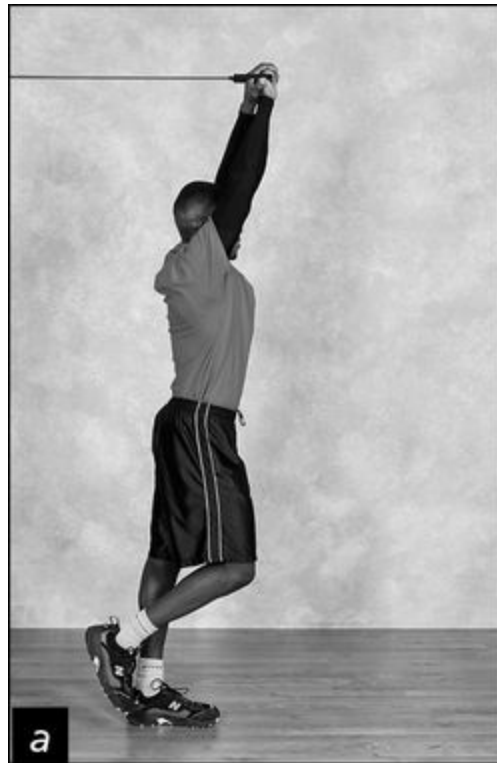
Side Bridge, Highest Level; Stir the Pot; and Overhead Cable Pull

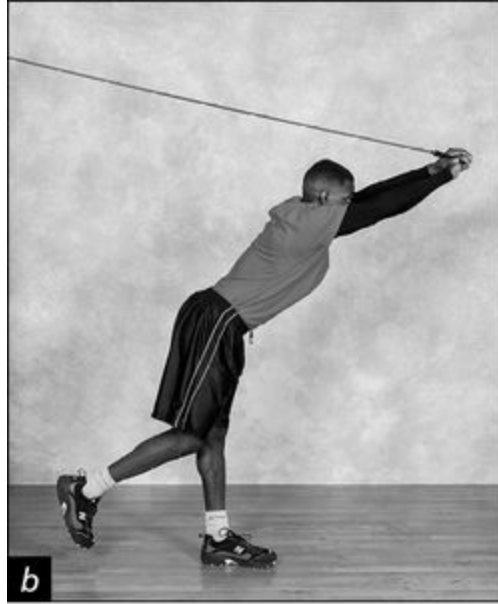
The side bridge can be made more challenging by incorporating the roll described in chapter 10, while also dynamically contracting and relaxing the abdominal wall while sustaining the position.



Stir the pot is an advanced core exercise for those who desire a longer endurance challenge. (a, b) Correct and (c) incorrect techniques (motion in the spine rather than containing motion about the shoulder joints) are shown here.

The patient or athlete can then begin abdominal work in a standing posture. A family of exercises that very cleverly trains the entire anterior chain comprises the overhead cable pull. Here the motion is focused about the hips; there is no spine motion (see photos).





The overhead cable pull is considered an athletic progression for abdominal exercise (a). However, it is more than simply an abdominal exercises because it enhances stiffness, strength, and control of the entire anterior chain. Note that all motion is about the hips (b).

Bird Dog, Highest Level

The bird dog is more demanding when the patient consciously stretches the hand out forward and the foot out behind—all the while ensuring that spine motion (particularly bending) does not take place. Enhancing the back and gluteal muscle is achieved by pushing the heel away and making a fist on the elevated limb side. **To enhance certain forms of performance** and if the patient has no exacerbation of symptoms, small wrist and ankle weights may be allowed. Plyometrics are added as the hand and foot draw opposing squares, as follows: The person begins in the starting position, and then moves the hand and the foot down and away from the body, and then downward. Just as the hand and foot are brought toward the body midline, the torso is consciously braced and a ballistic short-range extensor contraction occurs at the shoulder and hip. Once again, no motion must occur in the spine.





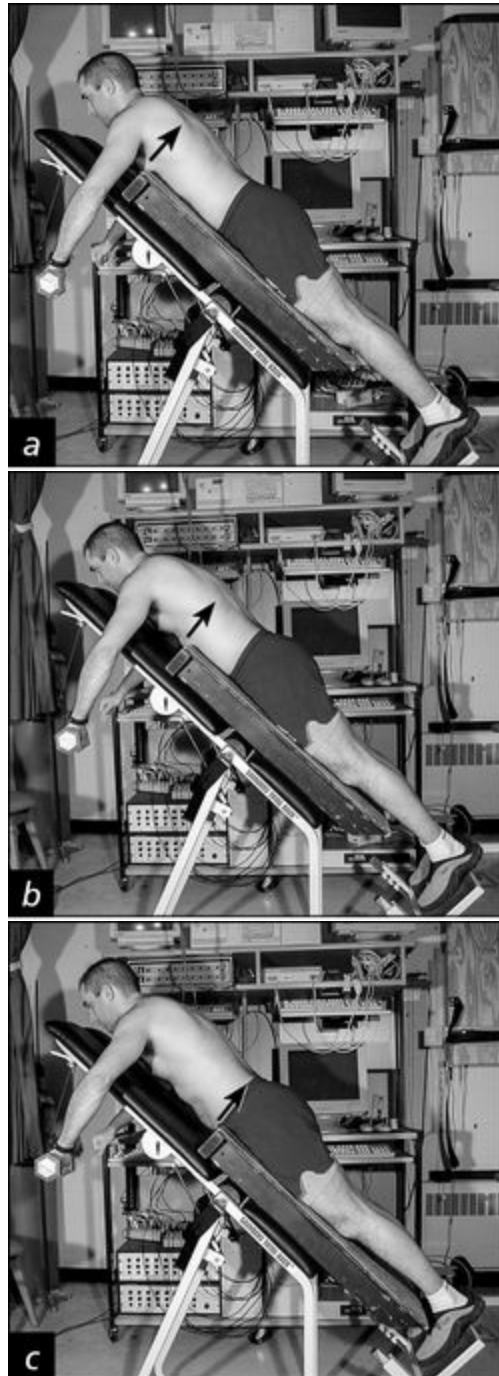


Educating All Back Extensor Muscle Motor Units

When searching for ways to obtain maximal myoelectric activity from the back extensor muscles, we discovered that isometric back extensor exercises do not recruit the full pool of motor units. Many more motor units fire with some extensor motion. As a result of this discovery, this back extensor exercise, which builds strength and endurance and helps to hypertrophy the muscles, was developed. It is used by some of the strongest athletes in the world to educate the motor units in the back musculature to fully recruit and fire.

While the athlete is lying prone on a bench with the torso supported on a movable stiff pad, place a weight in one hand. The edge of the pad is under the mid rib cage. The cantilevered portion of the spine is slightly flexed and then extended, combined with some slight twist back to neutral (a). The object is not to actually twist but to focus activation on one side of the extensors—to use mental imagery (chapter 10) to assist in activating the maximal number of available motor units. The spine never extends past neutral. After a set, place the weight in the athlete's other hand and have the athlete repeat the exercise. Then move the pad downward (perhaps about the level of the navel) so that a greater portion of the torso is cantilevered (b). Have the athlete repeat the sets. Move the pad farther down the athlete's body

(perhaps to about the pelvis), leaving more of the torso cantilevered, and have the athlete repeat the entire process (c). The sharp edge of the pad is better for imaging the edge and focusing on the corresponding section of muscle.



Photos from Stuart McGill

Squat With Overhead Resistance

Moving to a standing posture, the progression continues with a wonderful exercise that challenges the back and hip extensor mechanism. The squat with overhead, self-generated hand resistance grooves a squat pattern, with very robust recruitment of back and shoulder musculature for stability and stiffness.

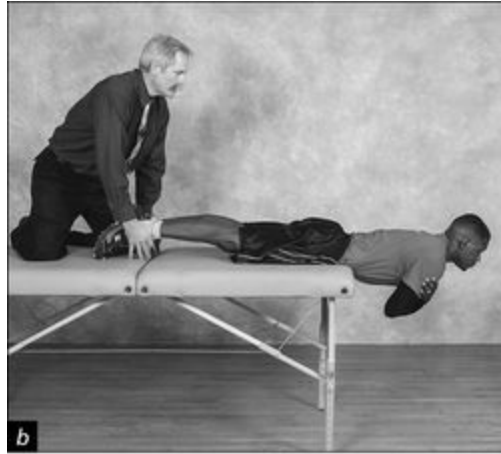
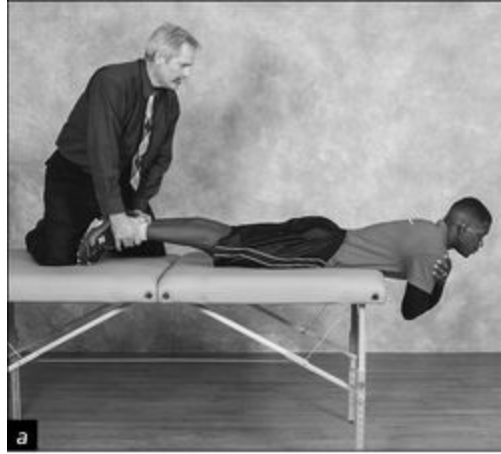




(a) The squat with overhead resistance requires one hand to push into the other causing contraction throughout the entire upper back, shoulders, and chest. (b) The squat uses the hip hinge, and the hips track along a 45° path (no spine motion).

Back Extensor Exercise Using a Bench or Table

Another variation of the exercise for educating all back extensor muscle motor units involves performing the extension motion over a table edge. The feet are secured, and the torso is progressively moved farther over the edge with each repetition (a, b).



Occupational and Athletic Work Hardening

Certain workers and athletes encounter demands that require training that we would never recommend to patients unless absolutely necessary. Anyone considering going beyond the exercises described previously should be fully aware of the increased risks of back reinjury that the exercises in this section pose. These exercises, which were developed for workers or occupational athletes, should be restricted to the groups they were developed for because they present a level of risk that is unacceptable for any but those willing to assume it for the sake of athletic or work performance.

Once again, choosing the appropriate challenge involves a blend of clinical art and science. Too little load will not produce the training adaptation, and too much will break tissue down. Listen to the body, and provide adequate rest.

Low Back Exercises for High-Performance Workers or Athletes

In general, we stage training of high-performance workers and athletes with the same steps used in our rehabilitation approach, except that two more steps are added. We begin in stage 1 by identifying the essential motions and establishing appropriate motion and motor patterns. Stage 2 is directed toward ensuring joint and whole-body stabilizing patterns. Stage 3 is to develop muscle endurance around these patterns. Stage 4 is directed to enhancing strength. And, finally, stage 5 is to establish power. Unfortunately, too many exceptional athletes are trained, or given a rehabilitation program based on strength and power, without an adequate foundation of stabilizing motion and motor patterns. They end up with back problems and are referred to us.

In this section we briefly consider exercises that may be helpful to either workers with demanding jobs or athletes. The requirements of the activities are not excessive for those who have adequately prepared by mastering all of the exercises previously discussed in the text. Specifically, these people should have mastered spine position awareness and be able to produce low back stabilizing patterns, but may need more work to move through the

ranges of motion specific to their tasks. In addition, they may need more strength and endurance.

Qualifying the Worker or Athlete

The clinical decisions involved in staging a worker or athlete through the progression of challenge is an art that can be assisted with scientific data. Those who are looking for a set recipe, however, will fail at creating a successful program for an individual. Generally, we blend the worker's or athlete's current exercise status and history of injury with our own knowledge of spine tissue loads that result from various activities and our knowledge of injury mechanisms to qualify someone for a specific exercise progression. We then put our educated guess to the test and monitor the person's progress to ensure the maintenance of a positive slope of improvement in both symptoms and function.

Lumbar Stability With Elevated Simultaneous Physiological Work Rates

Although some people can maintain spine stability over all sorts of activities, including those that require the stabilizing musculature to assist with physiological challenges such as elevated breathing, others cannot (McGill, Sharratt, and Seguin, 1995). We have noted a compromise in the ability to stabilize more often postinjury (McGill et al., 2003), although we have also detected that compromise in so-called virgin backs. Interestingly, tall athletes tend to be poorer in their ability to cocontract the abdominal muscles to ensure sufficient stability during high work rates and highly challenged breathing than their shorter counterparts. This perception appears to be supported by the most recent evidence showing that taller workers have a greater likelihood of having perturbed motor patterns while breathing heavily and holding spine loads (McGill et al., 2003).

Consider the warehouse worker, firefighter, or football player who must work at a high physiological rate that results in deep and elevated lung ventilation. The inability to maintain constant cocontraction in the abdominal wall (i.e., the muscles tend to relax during deep inhalation) is an indicator of compromised spine stability, particularly when heavy external loads that demand a stable spine are required. These people must develop stabilizing

motor patterns that transfer to all activities. Later in this section is an exercise we devised to train these people to maintain abdominal bracing while breathing heavily.

Push, Pull, Lift, Squat, Lunge, Carry, and Torsional Capability

Progressions in these movement patterns are essential for creating a balanced foundation for enhanced pain-free performance. Pulls usually begin with pull-ups. Emphasis is placed on a full and concentrated hand grip, concentrated scapulae, a prestiffened core, and so on. Better form, with fewer repetitions but more effort within the few repetitions, characterizes the beginning of each exercise progression. Push patterns generally begin with push-ups. For example, the bench press would never be recommended to someone who could not perform 20 fast and pristine push-ups. Lifts begin with the shortstop squat of weight elevated on blocks. One-legged kettlebell lifts may be added to the progression to challenge balance and precise spine control. Here the core is stiffened, grip effort is emphasized, and all motion occurs about the stance-leg hip. Kettlebell swings may be added for those with sufficient shear stability or stiffness.

Lunges may incorporate hand loads, but the person is screened for a sufficiently tight pelvic ring (e.g., a history of sacroiliac pain would disqualify the person). Carries may begin with light suitcase carries and progress to farmer's walks with appropriate coaching of core-stiffening techniques. Torsional exercises always begin with the ability to stop a twist. Athletes such as golfers, for example, greatly benefit from stop-twist exercises in both directions to enhance control and provide protection during the swing. Recall that generating torque about the twist axis imposes approximately four times the compression on the spine as an equal torque about the flexion–extension axis. This is why it is unwise to train for torsion generation until the back is quite robust and healthy.

As speed is added to the exercises, the ability to stop movement is as important as creating the movement. A balanced and successful program will address these considerations.

Figure 11.4 (a-c) Pull-ups are performed by first centrating the scapulae and enhancing the grip effort and core stiffness. Then the person pulses to accelerate the pull. (d-f) Conventional push-ups are performed with a neutral spine and a stiffened torso. The person pulses to accelerate the move. The staggered hand placement challenges control as the person ensures the torso does not laterally bend. Poor form is shown in photo f. (g-i) A one-legged good morning version of this exercise can be included for those interested in more hamstring challenge. The hands alternate between the right in front and the left in front. (j-l) Carrys begin in (j) suitcase style then progress to (k) racked and finally to (l) bottoms-up. (m-n) Lunges follow all of the spine-sparing principles by maintaining a neutral spine and motion occurring about the hip joints.

















Special Spine Exercises for Workers and Athletes

Although the exercises in this section have been designed to minimize injury risk, you must be careful to prescribe them only to those who can already safely perform the advanced levels of the big three exercises presented in chapter 10.

Establishing Spine Stability When Breathing Is Elevated

Some athletes and workers risk spine stability only when they are breathing heavily. To train simultaneous patterns to ensure stability during heavy breathing, we have developed the following approach:

1. Ride an exercise bike at an intensity to elevate ventilation.
2. Then immediately dismount and adopt the side bridge posture on the floor.

In this position the stabilizing musculature must remain isometrically contracted; otherwise, the bridge posture is lost. Heavy ventilation is also required, however, to groove the motor patterns that coordinate diaphragm contraction and the other thoracic muscles involved in efficient lung function. The curl-up and bird dog postures are also used after vigorous stationary biking to groove stable patterning in all of the supporting muscles. This establishes spine stabilization patterns in workers and athletes alike.

Suggestion for Those Whose Back Is Worse With Sitting and Then Have Trouble Standing

To combat the cumulative stresses of sitting, we often recommend walking over rough ground for an hour or two while wearing a backpack with a 5 to 10 kg (11 to 22 lb) load placed low in the pack.

Amazingly (to some), many patients report that this works wonders. The typical forward-flexed antalgic posture requires the back extensors to be active (which imposes a substantial load penalty on a flexed spine). The backpack acts as a counterweight to extend the spine and bring the torso upright. The back extensors are no longer needed to contract, effectively removing their contribution to spine load. Wearing the backpack reduces spine loading!

Low Back Exercises Only for Athletes

Specific athletic objectives require specific training techniques. As noted earlier, however, too many patients make the mistake of looking to athletes for training exercises under the misconception that the same approach will help their own backs. This critically important notion must be emphasized again: Athletic exercises are not for enhancing back health in patients. In addition, too many athletes use bodybuilding principles for building mass. These exercises must not be emulated by patients. The following exercise examples are reserved for athletes only.

Training-Specific Athletic Maneuvers

Trunk torsional machines and resisted twisting motions are reserved for those who want to excel at very special tasks such as Olympic discus

throwing. (Some track athletes with whom I have been associated clinically do not perform these exercises because they exacerbate their back symptoms.) Otherwise, the torsional moment capabilities are developed with the spine in a neutral position, with a fully braced torso musculature, but with a torsional moment challenge. The neutral position is the most robust posture for withstanding elevated spine loads; it is also the spine posture that is transferable to other activities that require torsional moments with the least risk of spine damage.

Sprinting and other power running events such as those performed by football players should be addressed here. Top-end speed for most sprinters is usually limited by the recovery of the leg in flexion (hip flexion), not a lack of hip extension power. Thus, many train the psoas according to the power philosophy—and end up with painful backs. It is critical for these athletes to up-train spine stability to support the added power and spine load during the rigorous hip flexor training required for sprinting events.

All sorts of exercises and special equipment have been developed for performance training. However, very few performance approaches have been documented with substantial scientific foundation to warrant discussion in a book such as this. However, one exception is the book *Supertraining*, in which the late Dr. Mel Siff synthesized and reviewed an impressive number of studies (Siff, 2003). For example, Dr. Siff made a case for avoiding the development of motions that are guided predominantly by machines, and instead advocated for the use of cables and free weights, as well as unloaded speed training, with data-based argument. Further, substantiated principles of functional anatomy, physiology, biomechanics, and motor aspects such as facilitation were used to justify the approaches documented in *Supertraining* to enhance strength, power, speed, and muscle endurance. These performance objectives cannot exist without the foundation of stability, mobility, and the additional principles documented in this book. Although there is no question that these principles and exercises are excellent for performance, there is a high resultant spine load. Some backs simply will not tolerate some of these higher-demand exercises—not everyone can tolerate the rigors of training needed to be a champion, regardless of the soundness of the scientific technique. The exercise variations are endless and are outside of the scope of this book.

Finally, with respect to designing workouts, the big three exercises can be performed at the beginning of the workout to establish the stable motor

patterns for the rest of the training session. These exercises also enhance core stiffness that lasts for a period of time (Lee and McGill, 2015) to enhance both safety and performance. Some trainers also like to finish the session with these exercises if they desire to take the exertions to fatigue.

Training to Squat and Power Clean

Squats and power cleans are good exercises for developing power for athletic performance, but they have limitations. Many top athletes are referred to us who have enormous sagittal plane strength, yet show great weakness in nonsagittal strength. They have overemphasized the Olympic lifts for power without training power for lateral motion, turns, and so on. Furthermore, these lifts create a lot of damaged backs. Many younger athletes do not have the hip structure to begin the pull with the hip flexion required to have a neutral spine at the beginning of the pull. Serious squat training for us always includes one-legged squats, the bowler's squat, stiff-legged sled drags, hip airplanes, and many more. This is a very involved topic that is discussed in more detail in *Ultimate Back Fitness and Performance* (McGill, 2014).

The instructions for performing squats and power cleans from our perspective are very simple: If a perfect starting position cannot be obtained, other means are needed. Many great athletes cannot do these lifts. Never sacrifice form for lifting more weight. Year after year I give this advice to young athletes, and year after year a substantial proportion of them ruin their backs by not heeding this simple guideline. As the old saying goes, It's amazing how much your parents learned as you grew older.

Very few people in North America can perform traditional squats and power cleans well, lifting the bar off the floor with a low risk of injury. The Eastern Europeans are technical masters, and the technique is well covered in McGill (2014). It is interesting to see the emphasis placed on establishing the correct motion and motor patterns long before substantial weight is attempted. Young Eastern European athletes spend years developing the form by lifting broomsticks. Only when form is perfect is strength increased and weight added to the bar. Generally, preserving the neutral lumbar spine will solve many of the safety issues, but this depends on the ability to take the shoulders and hips to extremes in the range of motion.

Some have found that placing the weight bar on blocks to raise the starting height improves the utility of this exercise for athletes other than

competitive weightlifters. This way a lifter can accomplish a fast lift without the larger back loads associated with the initial crouched posture needed to pull from the floor. In this power exercise, speed is important; participants are advised to train slow to be slow, train fast to be fast. If you are interested in these exercises, the actual weight kinematics for the full power clean and the coordination of the leg muscles, torso muscles, and those involved in the shoulder pull can be found in extensive detail in McGill (2014).

Performance exercise is so much more than lifting weights. In many cases the way the training programs are organized causes detriments to performance. For example, one high-performance athlete came for a consult wondering why, after training the squat with a bar on his back, he was able to squat more load but his vertical jump—which had been his original objective for training—had decreased. I had to show him that he had trained himself to be slow! Clearly, his objectives were not matched to the training approach.

If the objective is speed, the person should not endurance train or train slowly. Speed comes from cyclic ballistic contraction and relaxation. Thus, many athletes need to train speed of relaxation more than speed of contraction. Performance enhancements come from a refinement of the balance of qualities. A strong person who lacks balance and falls over when performing a one-legged squat may be fine in a weight room but a disaster on the football field. Strength, speed, and power all demand the foundation of appropriate motion and motor patterns, balance, and regional range of motion within the three-dimensional context of athletic motion. Training to achieve optimal performance is the topic of *Ultimate Back Fitness and Performance* (McGill, 2014), which is a companion text to this one for the real expert.

A Final Note

Rehabilitation approaches are continuing to embrace techniques that consider notions of torso stability and components involving posture, motor patterning, and appropriate progressive challenge. Many groups continue to work to do the following:

- Understand the contributions to stability of various components of the anatomy at particular joints—and the ideal ways to enhance their contributions.
- Understand the magnitudes and patterns of muscle activation required to achieve sufficient stability while sparing the joints.
- Identify the best methods to reeducate faulty motor control systems to both achieve sufficient stability and reduce the risk of inappropriate motor patterns occurring in the future.
- Develop motor patterns for optimal performance in athletes.

Collegial efforts between scientists and clinicians continue to develop the scientific foundation to justify better low back injury prevention and rehabilitation approaches. Much remains to be done.